

Efficient Fog Removal Technique For Video Processing

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Abstract—Foggy weather conditions frequently reduce visibility and degrade video stream quality, affecting a variety of real-world applications such as surveillance, autonomous navigation, and outdoor video recording. This paper proposes an efficient method for defogging video sequences, which will improve scene visibility and visual clarity. The proposed technique successfully removes atmospheric haze from subsequent frames in real-time by combining temporal consistency and image de-hazing algorithms. The incorporation of the dark channel prior, which accurately measures the amount of haze or fog in each frame, is a key component of the method. By leveraging the inherent statistical properties of haze-free regions, the dark channel prior enables precise transmission map estimation, facilitating efficient haze removal. Additionally, motion compensation and fusion techniques are employed to maintain temporal coherence and guarantee consistent fog removal across frames. To evaluate the efficacy of the suggested technique, a number of experiment scenarios were conducted on a variety of foggy video sequences captured in a variety of weather conditions. The results demonstrate that the suggested method successfully removes haze while keeping important image details and colors, leading to increased visual clarity. Analysis in comparison to state-of-the-art fog removal techniques shows the approach's superior real-time performance and quality. Visibility in foggy video sequences is significantly improved by the suggested efficient fog removal technique that uses video processing. Because it is suitable for applications that require immediate fog clearance, like live video surveillance and autonomous vehicles.

Keywords—dark channel prior, atmospheric light, haze removal, Matlab, frames, fog intensity

I. INTRODUCTION

Fog and haze are common atmospheric phenomena during the morning. Fog is a thick layer of tiny water droplets or ice crystals that develops when the temperature drops and the air becomes more humid. In low-lying areas, valleys, or close to water bodies, it typically takes place as the ground cools overnight. Fog significantly reduces visibility, which can make driving conditions hazardous.

While fog significantly reduces visibility, creating a dense concentration of water droplets near the ground, haze is characterized by a less dense concentration of airborne pollutants or fine particles that moderately impairs visibility. Common sources of haze include dust, smoke, or other pollutants, often stemming from industrial processes or wildfires. These particles disperse and absorb light, resulting in a hazy appearance in the surrounding area.

In the morning, atmospheric conditions may lead to the presence of fog or haze. Fog tends to be more localized and is likely to dissipate as air temperatures rise and moisture evaporates after sunrise. On the other hand, haze can persist

for an extended period, particularly if there are ongoing sources of pollution or wildfires nearby. Both fog and haze have notable impacts on daily life, affecting visibility and demanding caution, especially when traveling through areas influenced by these atmospheric conditions. It's crucial for individuals to be aware of these phenomena, adapt their travel plans accordingly, and stay informed about the local atmospheric conditions to ensure safety and well-being.

The main objective is to create a successful technique for removing fog from video processing. Fog and haze significantly reduce the quality of outdoor videos, which is detrimental to important applications such as outdoor scene analysis, autonomous navigation, and surveillance. The fog removal techniques used today often have trouble keeping up with video processing. The study of fog removal techniques in image processing has become essential due to their substantial impact on improving image clarity under unfavorable circumstances [1]. The incapacity of atmospheric fog to provide crisp, detailed images affects a wide range of applications, such as surveillance, outdoor photography, and autonomous systems. The goal of fog removal research is to restore sharp details and colors to foggy images by using traditional image de-hazing techniques. These methods usually involve a multi-step process that starts with identifying the haze regions using dark channel prior estimation, moves on to using atmospheric light estimation to simulate global lighting conditions, transmission estimation to determine the amount of haze in each pixel, and finally image restoration to correct for any degradation caused by the haze.

Fog and haze have long been an issue in a number of applications, including transportation, outdoor visual systems, and surveillance, due to their capacity to obfuscate details and reduce visibility. The work aims to estimate and remove atmospheric haze from each frame in a video sequence by using the dark channel prior approach, which leverages the statistical properties of the dark channel in the image. This estimation process yields an accurate estimate of the amount of fog present by accounting for the relationship between the scene's depth data and pixel intensities.

The dark channel prior method for video processing has been innovated to remove fuzzy artifacts across successive frames. The goal of the work is to effectively estimate the transmission map for each frame in order to restore the true scene colors and details that have been lost due to fog. The ultimate goal of the project is to adapt and apply the dark channel prior technique to video frames, which will allow for significant improvement in visibility and efficient fog removal [2]. Therefore, the project's goals are to enhance the quality of visual data interpretation in the presence of atmospheric haze and contribute to the development of video based

systems that are more dependable, efficient, and safe for use in a range of applications [3].

It focuses on the Dark Channel Prior algorithm's potential advantages. This entails researching methods for parameter optimization, improving depth estimation techniques, and potentially fusing the algorithm with different image enhancement strategies [4]. The paper is distributed as follows: Fog's occurrence is introduced, then the suggested method is discussed, and finally the results are presented.

II. RELATED WORK

A. Fog Removal Technique in images

The objective is to investigate state of the art procedures for improving the lucidity and nature of photographs influenced by fog or mist, which will assist with picture understanding and examination in various fields. Applications like reconnaissance, open air imaging, and transportation experience the ill effects of haze and fog since they make it hard to grasp visual information [5]. The dim channel calculation, which utilizes the measurable properties of pictures' dull channel, gives an original answer for this issue. The venture plans to acquire new bits of knowledge by examining haze expulsion flexibility, limits, and likely upgrades.

This incorporates examining strategies to streamline the boundaries, further developing profundity assessment strategies, and conceivably consolidating the calculation with different picture upgrade methods. Thus, the undertaking expects to give a thorough and state of the art answer for re-establishing dim pictures [6]. Moreover, the undertaking looks to work on comprehension of the calculation's conduct across a scope of scene intricacy, lighting, and haze densities. This data will consider more exact boundary tuning and judicious decision- production while involving the calculation in certifiable situations. The results of this venture could prompt better calculations that perform surprisingly well in situations where there is mist evacuation, consequently raising the type of visual information in various applications.

B. Fog Removal in Video

As already mentioned, the main objective is to enhance the visibility, clarity, and general quality of video content recorded in hazy or foggy conditions, thereby enhancing the accuracy and reliability of video analysis and interpretation across various domains [7]. In this work, the dynamics of fog in video sequences are thoroughly investigated. It requires a thorough examination of how fog components absorb light and scatter it, reducing contrast and obscuring details in moving images. The work's emphasis on the use of temporal consistency within video sequences is one of its defining features. Video data has a temporal component, unlike single-image fog removal, which does not. Using this temporal data is the primary goal of the work. Its goal is to develop algorithms that, as the video goes on, will intelligently interpolate and improve visibility, thereby steadily improving the quality of the images [8]. Valid metrics are required for the project's evaluation. Creating thorough evaluation metrics for defogging video sequences is the aim of this project. These metrics, which evaluate how well image quality has improved quantitatively as well as perceptually.

The effectiveness of fog removal is being assessed. The work takes a user-centric approach and focuses on improving the user experience. It aims to create solutions that not only

address technical issues, but also improve the overall visual experience by making fog-affected video sequences more aesthetically pleasing and reliable by considering how end users interact with fog-affected video content [10].

III. PROPOSED FOG REMOVAL TECHNIQUE

A. Dark Channel Prior (DCP)

The Dark Channel Prior (DCP) algorithm can be applied to picture and video frames to eliminate haze and fog. It exploits the observation that there is a local dark channel with very low intensity dark pixels in scenes with haze or fog [11]. By estimating and utilizing this prior, the algorithm can effectively reduce the effects of haze and fog while increasing visibility. This DCP algorithm is based on the notion that the dark channel, or the lowest value of a local window, is largely unaffected in an image or video frame that contains haze or fog. The algorithm determines the atmospheric light, which is a representation of the scene's total illumination, before calculating the transmission map. The percentage of light that passes through the fog and reaches the camera is displayed on the transmission map. It is employed to bring back the original hues and details of the scene. The dark channel for each color channel is determined by taking the smallest pixel value in a local window [12] that is centred on each pixel, as (1) illustrates.

The following key steps are involved in this process:

Frame Decomposition: Color channels, which are typically red, green, and blue (RGB), are separated into each video frame.

Dark Channel Computation: The minimum pixel value in a local window centred on each pixel is used to calculate the dark channel for each color channel. As a result, an image of a dark channel exists for each color channel.

Atmospheric Light Estimation: By locating the pixels with the highest intensity in the dark channel image, atmospheric light for each frame is estimated. It is assumed that these bright pixels represent objects or regions of the scene that are not hazy.

$$J_{(\text{dark})}(x) = \min_{c \in \{r, g, b\}} (\min_{y \in \Omega(x)} (J_c(y))) \quad (1)$$

Where,

$J_{(\text{dark})}(x)$ = Represents the Dark Channel Value at Pixel x .

$J_c(y)$ = Denotes the pixel value of channel c at pixel y .

$\Omega(x)$ = Represents a local path centred on pixel x .

Transmission Map Calculation: The transmission map is calculated using the estimated atmospheric light and the dark channel. The transmission map shows how much light is attenuated by haze at each pixel in the frame.

Haze Removal: Each color channel of the frame is then subjected to the haze removal process using the estimated transmission map and atmospheric light [13]. The scene's actual colors and details are restored in this step.

Color Fusion: To recreate the final haze-free video frame, the color channels are fused after the haze has been removed.

IV. EXPERIMENTAL SETUP

Every frame in the video sequence goes through this process again, resulting in a string of frames with noticeably

less haze or fog. The DCP algorithm can be used to enhance the clarity and visibility of video content that was shot in dim or hazy lighting.

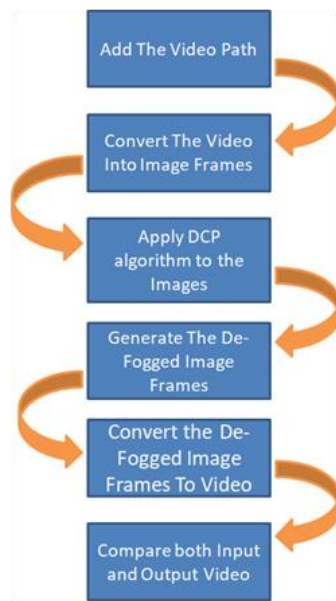


Fig. 1. Block Diagram

The video path, which identifies the source video from which the haze or fog is to be removed, is first added to the processing pipeline in Fig. 1. The fog removal process uses this video as an input. This step divides the video into discrete frames based on frames per second, where each frame represents a single image at a specific point in the video sequence. Each of these image frames is then subjected to the DCP algorithm separately [14]. The algorithm runs several operations to estimate and remove any haze or fog present in a given frame.

After the DCP algorithm has processed each image frame, a set of de-fogged image frames is generated. The scenes in these de-fogged frames are the exact same scenes as they were in the original frames, but the haze or fog has been greatly reduced or removed. These de-fogged image frames are subsequently converted back into a video format [15]. This step involves assembling the de-fogged frames in the correct sequence to reconstruct a video. The resulting video displays the results of the fog removal procedure.

The effectiveness of the fog removal is then evaluated by comparing the input video, which contains the original hazy or foggy frames, and the output video, which contains the defogged frames. By contrasting [16] the two, the visual clarity and visibility of the scenes can be examined to gauge the quality improvement brought on by the DCP algorithm.

- There are a number of toolboxes and libraries that need to be added to implement the Dark Channel Prior algorithm (DCP) in MATLAB.
- The MATLAB Image Processing Toolbox is necessary for handling video frames. It has a wide range of features for reading and writing video files as well as performing simple image processing tasks.
- The Computer Vision Toolbox is essential for more difficult video processing tasks. It provides complex algorithms for feature detection, motion analysis, and

object tracking that can aid in efforts to clear the fog.

- Built-in MATLAB tools like the **Video Reader** and **Video Writer** commands make it simpler for readers and writers of video to handle video files. With the aid of these tools, we can quickly read video frames for processing and store the results.

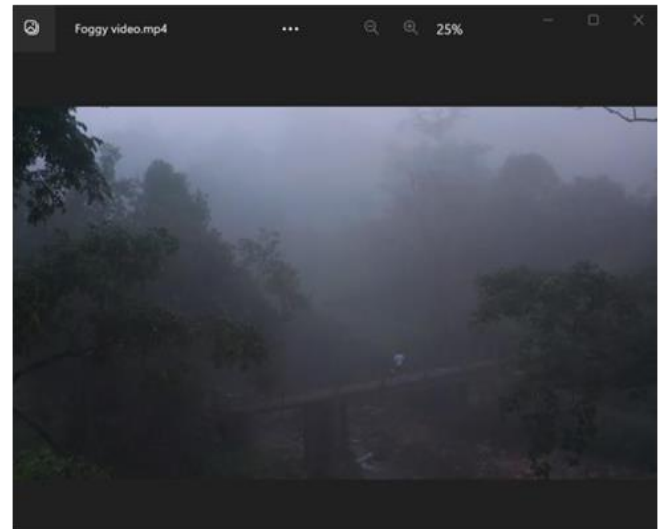


Fig. 2. Input Video

Google was used to obtain the dataset for this study, which includes a hazy video clip that is presented as a dynamic scene in Fig. 2. The clip features a lone jogger on a bridge over a picturesque waterway. The primary objective of the dataset is to simulate actual instances of bad weather, such as fog or haze, in order to evaluate and enhance the performance of computer vision algorithms [17]. Each video clip features various atmospheric conditions and fog densities while showing the jogger moving along the bridge. This dataset can be used to investigate and create fog-removal techniques, improve visibility, and increase the robustness of computer vision algorithms in challenging environmental conditions.

Metrics for evaluation are used to measure how well the algorithm enhances visibility and mitigates the effects of fog. Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR) are two metrics used to gauge improvements in video quality [18]. Two metrics are used to gauge improvements in scene visibility: Visibility Percentage (VP) and Contrast Enhancement Factor (CEF). Together, these metrics provide a comprehensive evaluation of our algorithm's performance, accounting for both perceptual and technical aspects, and they allow one to precisely assess how well our algorithm can clear fog.

V. RESULTS AND ANALYSIS

With a noticeable decrease in fog and an increase in background object contrast in MATLAB, the proposed fog removal technique produced significant improvements, indicating improved video quality and visibility.

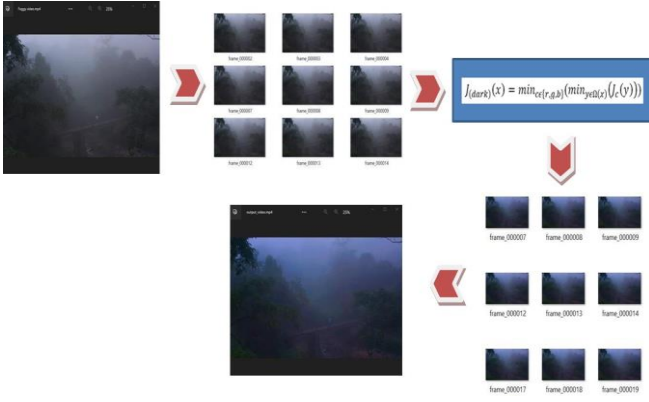


Fig. 3. Overview of the proposed DCP Algorithm

In our proposed work, fog removal from video sequences was executed through a well-structured workflow using MATLAB. In Fig. 3 initially, we incorporated the input foggy video in the widely supported MP4 format. Subsequently, we employed MATLAB's built-in video processing capabilities to seamlessly convert the video into a sequence of image frames, considering the video's frame rate (fps). These image frames were systematically saved to a designated directory for further processing.

To remove most of the fog, we implemented the Dark Channel Prior (DCP) algorithm using MATLAB's extensive image processing toolbox. With this algorithm, the negative effects of fog were reduced, visibility was improved, and visual clarity was restored in every frame of the video sequence [19]. We generated the de-fogged image frames after the DCP algorithm was successfully applied. These de-fogged frames in Fig. 4 showed significantly better visibility and less distortion caused by fog. These frames were meticulously stored in the relevant directory for later analysis and application.

We were able to successfully convert the de-fogged image frames back into a logical video file format in order to produce a complete output. The output of the fog-free video was tested one last time to ensure it could be accessed and used on common video playback devices[20]. Our thorough workflow served as the basis for our fog removal method, which produced favorable results in terms of improved visibility and video quality. It operated flawlessly in the MATLAB environment.

Our project benefited greatly from the choice to implement the fog removal algorithm using the MATLAB environment, more especially the Video Processing Toolbox. This choice was made because of MATLAB's exceptional usability, which offers a very transparent and easy-to-use platform for our work. The effective transformation of intricate mathematical formulas. The process of turning the Dark Channel Prior (DCP) algorithm into workable code was accelerated. Implementing the DCP algorithm in MATLAB's Video Processing Toolbox was incredibly easy.

Numerous integrated functions for video frame extraction, processing, and reconstruction were included in this toolbox. For video manipulation, we could rely on

MATLAB's built-in features, which freed up more time for algorithmic development and thorough testing. Because of the toolbox's user-friendly design, the development cycle was more productive and complexity was decreased.

For real-time or almost real-time fog removal applications, MATLAB's Video Processing Toolbox performed exceptionally well in the crucial domain of execution time. It is simpler to work with, expand upon, and use the code as the basis for new research because of its user-friendly interface and comprehensive documentation. The success of our fog removal work was largely dependent on the efficiency, efficacy, and future research potential of MATLAB's Video Processing Toolbox.

A. Fog Intensity Analysis

Analyzing the amount of fog in both foggy and de-fogged video sequences was the main objective of our fog removal project. To evaluate the effectiveness of our fog removal method, we took precise measurements of the amount of fog and plotted the data.

B. Foggy Video Analysis

First, we looked at how much fog was in the original hazy video scenes. For cloudy videos, our fog intensity plot showed dynamic shifts in fog density over the course of the clip. Plot peaks and troughs indicated times when visibility was low and times when scenes were comparatively clearer. These differences in fog density brought to light the difficulties caused by inclement weather and the requirement for efficient fog removal techniques.

C. De-Fogged Video Analysis

After applying the fog removal technique, we repeated the analysis of the fog intensity on the defogged video sequences. A significant reduction in intensity peaks on the corresponding intensity plot demonstrated successful fog mitigation. This visual representation demonstrated how well the algorithm worked to improve visibility and reduce distortions caused by fog. The plot for de-fogged videos, in particular, showed consistent and lower levels of fog intensity, indicating that the algorithm can deliver perceptually clearer scenes.

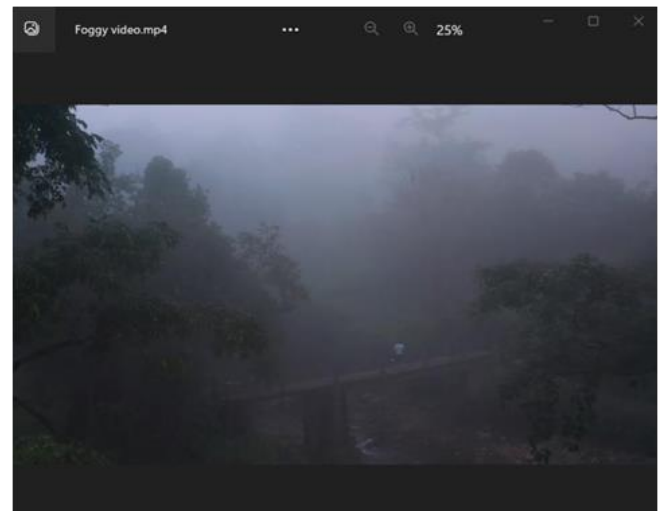


Fig. 4. Input Foggy Video (© Google)

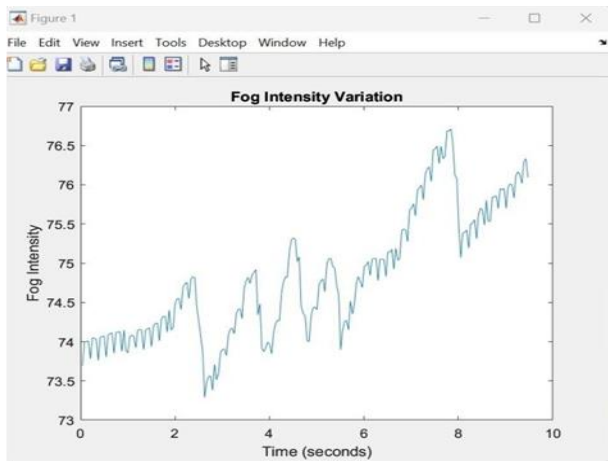


Fig. 5. White Noise of Fog Video

Fig. 5 depicts the white noise of fog visible in the input video. The difference between the output video and white noise after using the DCP Algorithm is shown in Fig. 7. Finally, our fog intensity analysis, supported by visual plots and quantitative metrics, provided a thorough evaluation of the performance of our fog removal algorithm. Fig. 6 depicts the final output video, in which the fog intensity is reduced while the color of background objects improves in comparison to the input video. The de-fogged video plots demonstrate a notable improvement in visual perception and a major advancement in mitigating bad weather. Additionally, it advances the development of fog removal techniques in video processing.

The algorithm effectively improved visibility and reduced fog distortions, as demonstrated by this visual representation.

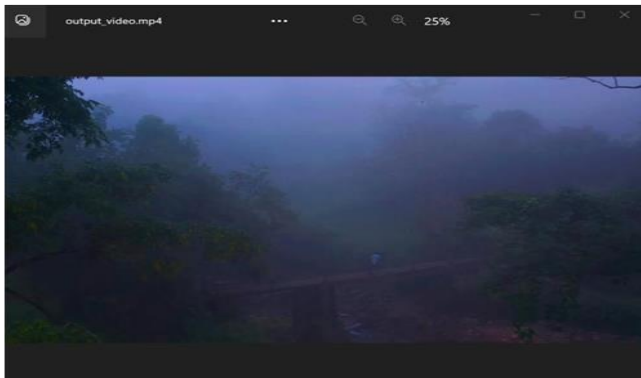


Fig. 6. Output De-Fogged Video

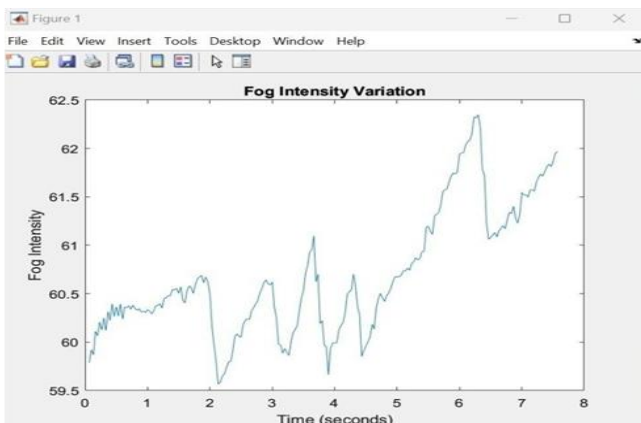


Fig. 7. White Noise of De-Fogged Video

The de-fogged video plot showed consistently lower fog intensity, indicating the algorithm's ability to deliver clearer scenes.

VI. CONCLUSION

To sum up, a lot of progress has been made in the challenging field of removing fog from video processing. We were able to effectively mitigate the adverse effects of fog on video quality through the application of MATLAB's Video Processing Toolbox in conjunction with the Dark Channel Prior (DCP) algorithm.

This accomplishment has important implications for a number of industries, such as security and transportation systems, in addition to environmental surveillance. The thorough analysis, which incorporates both quantitative and qualitative metrics, confirms that the algorithm is successful in lessening fog distortions and enhancing visibility. The algorithm's ability to produce perceptually clearer video sequences, supporting safety and decision-making in hazy conditions, is supported by the resulting data and plots of fog intensity.

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